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APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

Second Semester B.Tech Degree (R, S) Examination May 2024 (2019 Scheme)

Course Code: MAT 102**Course Name: VECTOR CALCULUS, DIFFERENTIAL EQUATIONS AND TRANSFORMS
(2019 SCHEME)**

Max. Marks: 100

Duration: 3 Hours

PART A*Answer all Questions. Each question carries 3 Marks*

Marks

- 1 If $\vec{r} = e^{-t} \hat{i} + e^t \hat{j}$ is the position vector of a moving particle, find its velocity at $t = 0$. (3)
- 2 Find a unit vector in the direction in which $f(x, y) = 4x^3y$ increases most rapidly at $P(-1, 1)$, and find the rate of change of f at P in that direction. (3)
- 3 Evaluate $\int_C (x^2 - 3y)dx + 3x dy$, using Green's theorem, C being the circle $x^2 + y^2 = 4$. (3)
- 4 Determine whether the vector field $\vec{F}(x, y, z) = x^3 \hat{i} + y^3 \hat{j} + z^3 \hat{k}$ is free of sources and sinks. If it is not, locate them. (3)
- 5 Find whether the solution set $\{x \sin x, x \cos x\}$ forms a basis or not. (3)
- 6 Solve $y''' + y'' = 0$. (3)
- 7 Find the Laplace Transform of $e^{-2t} \sin 5t$. (3)
- 8 Find the inverse Laplace Transform of $\frac{4}{(s+1)^4}$. (3)
- 9 Find the Fourier cosine integral of $f(x) = \begin{cases} 1, & \text{if } |x| < 1, \\ 0, & \text{if } |x| > 1 \end{cases}$ (3)
- 10 Find the Fourier sine transform of e^{-x} . (3)

PART B*Answer one full question from each module, each question carries 14 marks***Module I**

- 11 a If $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ and $r = |\vec{r}|$, prove that $\nabla f(r) = \frac{f'(r)}{r} \vec{r}$. (7)
- b Prove that the line integral $\int_{(-1,2)}^{(0,1)} (3x - y + 1)dx - (x + 6y + 2)dy$ is independent of the path. Also find its value. (7)

OR

- 12 a Find the work done by the force field $\vec{F}(x, y, z) = z \hat{i} + x \hat{j} + y \hat{k}$, where C is the curve $\vec{r}(t) = \sin t \hat{i} + 4 \sin t \hat{j} + \sin^2 t \hat{k}$, $0 \leq t \leq \pi/2$. (7)

- b Find $\nabla \times (\nabla \times \vec{f})$, if $\vec{f} = y^2 x \hat{i} - 3yz \hat{j} + xz \hat{k}$ (7)

Module II

- 13 a Find the area of the ellipse $x = a \cos t$, $y = b \sin t$; $0 \leq t \leq 2\pi$ using line integrals. (7)

- b Use Stoke's theorem to evaluate $\int_C \vec{F} \cdot d\vec{r}$, where $\vec{F}(x, y, z) = xy \hat{i} + x^2 \hat{j} + z^2 \hat{k}$ and C is the intersection of the rectangle $0 \leq x \leq 1$, $0 \leq y \leq 3$, and the plane $z = y$. (7)

OR

- 14 a Using Green's theorem evaluate $\int_C x \cos y \, dx - y \sin x \, dy$ where C is the square with vertices $(0, 0)$, $(0, \pi)$, (π, π) , and $(\pi, 0)$. (7)

- b If $\vec{F} = x^3 \hat{i} + y^3 \hat{j} + z^3 \hat{k}$, σ is the surface of the cylinder bounded by $x^2 + y^2 = 4$, $z = 0$, $z = 4$, find the outward flux of $\vec{F}$ across σ using Divergence theorem. (7)

Module III

- 15 a Solve the initial value problem $(D^2 + 4D + 5)y = 0$, $y(0) = 2$, $y'(0) = -5$. (7)

- b Solve $y'' - 6y' + 9y = \frac{e^{3x}}{x^2}$, by the method of variation of parameters. (7)

OR

- 16 a Solve by the method of undetermined coefficients, $y'' - 4y' + 3y = \sin 3x$. (7)

- b Solve $x^2 y'' + 7xy' + 13y = 0$, $y(1) = 0$, $y'(1) = 2$. (7)

Module IV

- 17 a Solve the differential equation using Laplace transform, $y'' + 2y' + 6y = 6te^{-t}$, given that $y(0) = 2$, $y'(0) = 5$. (7)

- b Find the inverse Laplace transform of (i) $\frac{2s+1}{s^2+2s+5}$ (ii) $\frac{e^{-s}}{s^2+2s+1}$. (7)

OR

- 18 a Using convolution find the inverse Laplace transform of $\frac{1}{s^2(s^2+a^2)}$. (7)

- b Find the Laplace transform of (i) $\sin^2 3t$ (ii) t^2 in $1 \leq t \leq 2$. (7)

Module V

- 19 a Using Fourier integrals prove that (7)

$$\int_0^{\infty} \frac{\cos\left(\frac{\pi\omega}{2}\right)}{1-\omega^2} \cos \omega x d\omega = \begin{cases} \frac{\pi}{2} \cos x, & |x| < \pi/2 \\ 0, & |x| > \pi/2 \end{cases}$$

- b Find the Fourier Cosine transform of $f(x) = \begin{cases} x, & \text{if } 0 < x < 2, \\ 0, & \text{if } x > 2 \end{cases}$ (7)

OR

- 20 a Find the Fourier sine integrals of $f(x) = \begin{cases} \pi - x, & \text{if } 0 < x < \pi, \\ 0, & \text{if } x > \pi \end{cases}$ (7)

- b Find the Fourier sine transform of $f(x) = \begin{cases} 1, & \text{if } 0 < x < 1, \\ 0, & \text{if } x > 1 \end{cases}$ (7)

Hence deduce that $\int_0^{\infty} \frac{1-\cos \omega}{\omega} \sin\left(\frac{\omega}{2}\right) d\omega = \begin{cases} \frac{\pi}{2}, & \text{if } 0 < x < 1, \\ 0, & \text{if } x > 1 \end{cases}$
